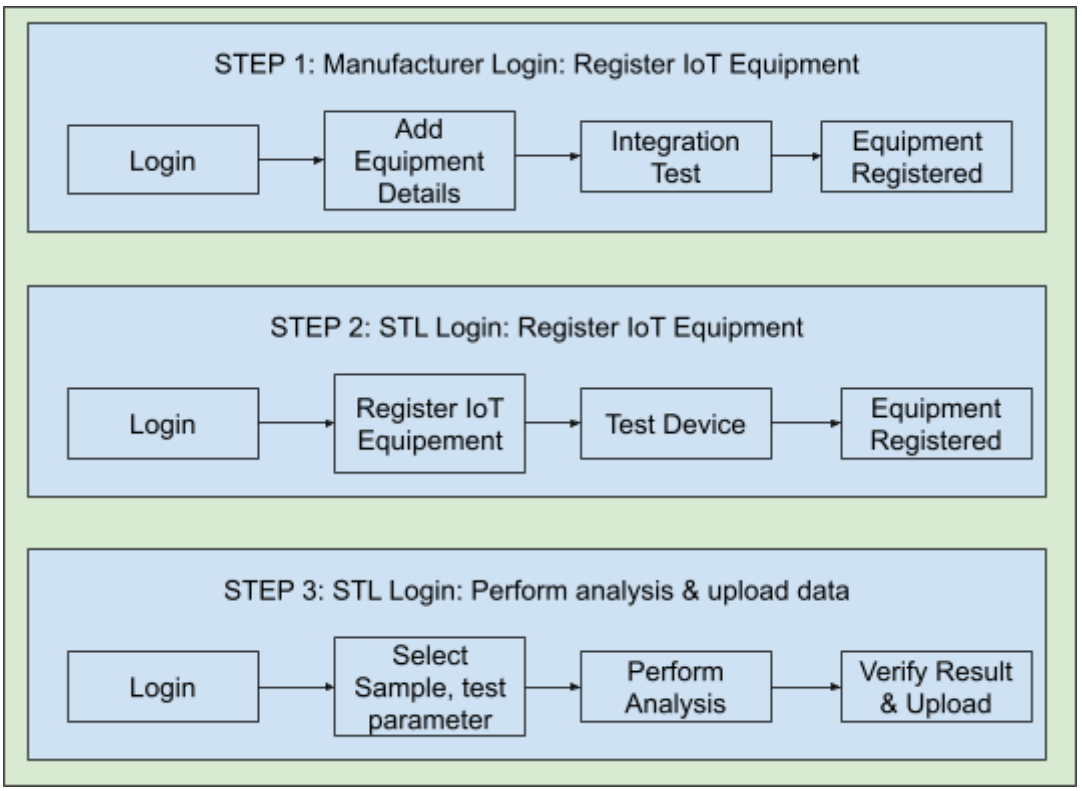
IoT Integration with SHC Portal

IoT Integration Methodology



Step I: Equipment Model registration on the portal

1. **Manufacturer's Login**: Device manufacturers will be provided with a login account to register a model. This login will allow device manufacturers to register their equipment models, get reports of the total equipment registered, and count the total devices connected to the SHC portal using IoT.
2. **Model Registration:** To register a model, the manufacturer will need the following details: model number, test parameters, interface type (USB/Serial/Bluetooth), and string format.
3. **Testing of Integration:** After registration, the manufacturer needs to test the integration by connecting the equipment and sending test data. If integration is successful, then registration is complete.
4. **Subsequent Steps:** Once registration of an equipment model is successful, soil testing laboratories can register their device on the portal for IoT integration.

Step II: IoT Device registration by Soil testing Laboratories (STL)

1. **Login to the Portal:** Soil testing laboratories (STLs) need to log in to the Soil Health Card (SHC) portal.

2. **Connect Soil testing Equipment:** Connect the microprocessor-based lab equipment to an internet-connected computer using a Serial interface cable.
3. **Register Device:**
    a. Navigate to the **IoT Devices** section.
    b. Select **Add Device** and enter the following details:
        - **Interface Type:** COM Port, App based(requires app integration)
        - **Device Manufacturer**
        - **Test Parameter(s)--** multiple allowed
        - **Model**
        - **Result Format:** Absolute values /Raw (needs further calculation)
    c. Click on the **Proceed** button.
4. **Device Data Reception:** (if COM/USB connected devices)
    - The screen will prompt you to press the **PRINT** button on the equipment.
    - Once the data is sent by the equipment, the screen will display the received data along with the test values.
    - If the data received is in Raw format, the user will configure the factors that are required for calculation. This is to be done only during registration. The option to modify these parameters should be given during test result uploading.
5. **Confirmation of Registration:** If the data is received and displayed correctly, the device registration is considered successful.
6. **Mobile app integration:** If the device uses a Mobile app to operate the device, in such a case mobile app integration will be done by the SHC mobile app team.

Step III: Soil analysis with IoT-enabled test equipment

1. **Sample and Parameter Selection:**
    a. Log in to STL account.
    b. Select the soil sample to be tested and choose the test parameter.
2. **Device and Test Information:**
    a. Based on the IoT registration data, the screen will display the manufacturer and model name of the device associated with the selected parameter.
    b. Perform the test using the indicated device.
3. **Data Retrieval:**
    a. After completing the test, click the **“Get Data”** button and select the relevant COM port.
    b. The screen will prompt you to press the **“Print”** button on the IoT-enabled lab equipment.
4. **Data Display and Verification:**
    a. Once the data is received, it will be displayed on the screen along with farmer details.
    b. If the device is configured to test multiple parameters, the user will select the test parameter here.
    c. If data is received in Raw format, the application will calculate the result based on factors configured during the device registration. User can modify the factors if needed.
    d. Verify the results.
5. **Save Results:** Press the **“Save”** button to store the test results on the SHC server.

Proceed to Next Sample: User can now proceed with testing the next soil sample.

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